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Auditors' Intention to Engage in Sustainability Assurance: Evidence From Vietnam

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Abstract: This study examines factors associated with external auditors' intention to engage in sustainability assurance in Vietnam, where sustainability reporting is expanding while assurance practice remains uneven. The research model is informed by institutional theory, self-efficacy theory, and the attitude-intention relationship in the Theory of Planned Behavior (TPB). Survey data from 317 external auditors working in independent audit firms were analyzed using partial least squares structural equation modeling (PLS-SEM). The results show that perceived regulatory and institutional pressure, perceived sustainability literacy, and sustainability assurance self-efficacy are positively associated with attitude toward sustainability assurance, whereas professional skepticism is not significantly associated with attitude. Perceived regulatory and institutional pressure, sustainability assurance self-efficacy, and attitude are positively associated with intention to engage. Mediation analysis shows significant indirect relationships through attitude for perceived regulatory and institutional pressure, perceived sustainability literacy, and sustainability assurance self-efficacy, but not for professional skepticism. The study extends sustainability assurance research by examining auditors' individual-level readiness and distinguishing how institutional signals, perceived sustainability literacy, task-specific self-efficacy, and attitude relate to readiness to engage in sustainability assurance. The findings highlight the professional capacity needed to support the development of credible sustainability reporting and assurance in an emerging economy.

Keywords: Sustainability assurance; External auditors; Behavioral intention; Attitude; Institutional pressure; Perceived sustainability literacy; Professional skepticism; Self-efficacy

1. Introduction

Sustainability assurance has become increasingly important as companies face growing expectations to provide credible information on environmental, social, and governance (ESG), and climate-related matters. Sustainability reports and ESG disclosures are increasingly used by investors, regulators, and other stakeholders to assess corporate risks, accountability, and sustainability performance. However, these disclosures often contain qualitative narratives, estimates, forward-looking statements, and non-financial indicators that can be difficult to verify and compare. These characteristics raise concerns about information credibility and the risk that sustainability claims may not be adequately supported. Prior research suggests that independent assurance can strengthen users' confidence in sustainability information and improve the perceived credibility of sustainability disclosures (Ballou et al., 2018; Hodge et al., 2009; Moroney et al., 2012; Pflugrath et al., 2011).

The relevance of sustainability assurance extends beyond the auditing profession. Credible assurance can support the implementation of sustainability reporting requirements, strengthen accountability, and improve trust in information used by stakeholders to evaluate sustainability performance. Earlier research has also highlighted the role of assurance in enhancing the credibility and legitimacy of sustainability reporting practices (Cohen & Simnett, 2015; O'Dwyer & Owen, 2005; O'Dwyer et al., 2011). Recent regulatory and professional developments reinforce this role. The issuance of International Standard on Sustainability Assurance 5000 (ISSA 5000) provides a global framework for sustainability assurance engagements, while related ethical and independence standards clarify the responsibilities of assurance practitioners (International Auditing & Assurance Standards Board, 2024; International Ethics Standards Board for Accountants, 2025). These developments indicate that sustainability

assurance is becoming a more formalized component of sustainability reporting and governance. Effective implementation, however, also depends on professionals who are willing and able to perform this work.

Vietnam provides a relevant setting for examining this issue because sustainability reporting is advancing while external assurance remains uneven. Sustainability-related disclosure is no longer entirely voluntary. Listed and public companies are subject to sustainability and environmental disclosure requirements under Circular No. 96/2020/TT-BTC (Ministry of Finance, 2020), while selected high-emitting facilities are required to conduct greenhouse gas inventories under Decision No. 13/2024/QD-TTg (Prime Minister of Vietnam, 2024). Evidence from leading listed companies also suggests a substantial gap between sustainability reporting activity and external assurance. Among the top listed companies examined in Vietnam, only 10% disclosed external assurance, compared with 49% across the wider Asia-Pacific sample (PwC Vietnam, 2023). More recent evidence indicates that ESG reporting activity has continued to expand. In PwC Vietnam's 2025 survey, 57% of respondents reported external ESG reporting, although many organizations still lacked structured reporting approaches and mature ESG data capabilities (PwC Vietnam, 2025).

The Vietnamese context therefore illustrates a broader challenge in the development of sustainability assurance. As sustainability reporting expands, the credibility of reported information depends not only on disclosure requirements and corporate reporting systems but also on the professional capacity available to evaluate that information independently. Exposure to sustainability assurance engagements, international standards, and specialized expertise may differ across the audit market. Public evidence on how sustainability assurance capability is distributed across Big 4 and local audit firms in Vietnam remains limited. Recent evidence from other ASEAN emerging markets similarly shows that sustainability assurance remains uneven and continues to depend on institutional development, reporting maturity, and assurance capacity (Handayati et al., 2026; Kurniawan & Basuki, 2024).

Despite growing academic attention to sustainability assurance, much of the existing literature has focused on firm-level assurance decisions, assurance provider choice, assurance quality, and the effects of assurance on reporting credibility or users' judgments (Ballou et al., 2018; Bentley-Goode et al., 2025; Casey & Grenier, 2015; Gipper et al., 2025; Li et al., 2025; Simnett et al., 2009). This literature has improved understanding of why firms obtain assurance and what assurance may achieve. Less attention has been given to the individual auditors who may be expected to perform sustainability assurance engagements. This represents an important gap because standards, reporting requirements, and client demand cannot by themselves create effective assurance capacity. The development of sustainability assurance also depends on whether auditors perceive this work as relevant, feel capable of performing it, and intend to participate when opportunities arise.

Behavioral intention is particularly relevant in a setting where actual sustainability assurance opportunities are not yet equally available to all auditors. Intention reflects an individual's motivational readiness to perform a behavior and is closely related to the individual's evaluation of that behavior (Ajzen, 1991; Ajzen, 2020). Recent research on sustainability-related regulatory transitions also suggests that accounting professionals' responses are shaped by both institutional conditions and individual-level factors (Krasodomska et al., 2025). Accordingly, this study examines auditors' intention to engage in sustainability assurance in relation to perceived regulatory and institutional pressure, perceived sustainability literacy, professional skepticism, sustainability assurance self-efficacy, and attitude toward sustainability assurance.

The research model is informed by three complementary theoretical perspectives. Institutional theory explains how regulatory, professional, and stakeholder pressures may shape auditors' attitude toward sustainability assurance and their intention to engage in it (Campbell, 2007; DiMaggio & Powell, 1983; Latif et al., 2020; Martínez-Ferrero & García-Sánchez, 2017). Self-efficacy theory explains why auditors' perceived capability to perform sustainability assurance tasks may influence both their attitude and intention (Bandura, 1977; Chen et al., 2001). The Theory of Planned Behavior (TPB) informs the relationship between attitude and behavioral intention (Ajzen, 1991; Ajzen, 2020). The study does not test the full TPB model. Subjective norms and perceived behavioral control are not modeled as TPB constructs because the framework separately examines external institutional signals and task-specific perceived capability through institutional theory and self-efficacy theory. Perceived sustainability literacy and professional skepticism are included as individual attributes that may further shape how auditors evaluate sustainability assurance. The resulting framework therefore links external institutional signals, individual perceptions and professional attributes, attitude, and behavioral intention within a single model.

Accordingly, this study addresses the following research questions:

RQ1. How are perceived regulatory and institutional pressure, perceived sustainability literacy, professional skepticism, and sustainability assurance self-efficacy associated with external auditors' attitude toward sustainability assurance?

RQ2. How are perceived regulatory and institutional pressure, sustainability assurance self-efficacy, and attitude toward sustainability assurance associated with external auditors' intention to engage in sustainability assurance?

RQ3. Does attitude toward sustainability assurance mediate the relationships between the proposed antecedents and external auditors' intention to engage in sustainability assurance?

To address these questions, this study uses survey data from 317 external auditors working in independent audit

firms in Vietnam. The proposed relationships are tested using partial least squares structural equation modeling (PLS-SEM). The analysis focuses on six constructs: perceived regulatory and institutional pressure, perceived sustainability literacy, professional skepticism, sustainability assurance self-efficacy, attitude toward sustainability assurance, and intention to engage in sustainability assurance.

This study makes three contributions. First, it extends sustainability assurance research by examining auditors' readiness as an individual-level component of the professional capacity required to support credible sustainability reporting. This perspective connects auditors' intention with the broader development of sustainability assurance infrastructure rather than treating assurance solely as a firm-level reporting decision. Second, the study develops an integrated framework that distinguishes external institutional signals, perceived sustainability literacy, professional skepticism, task-specific self-efficacy, attitude, and intention. This framework helps explain how institutional conditions and individual-level factors may jointly shape auditors' readiness to participate in sustainability assurance. Third, the study provides evidence from Vietnam, where sustainability reporting requirements and ESG practices are expanding but external assurance remains uneven. The findings therefore contribute evidence from an emerging economy undergoing a transition toward more formalized sustainability reporting and assurance practices.

The remainder of the paper is structured as follows. Section 2 develops the theoretical framework and hypotheses. Section 3 describes the research methodology. Section 4 reports the empirical results. Section 5 discusses the findings, implications, limitations, and directions for future research. Section 6 concludes the study.

2. Theoretical Background and Hypothesis Development

2.1 Sustainability Assurance and Auditors' Behavioral Intention

Sustainability assurance is becoming an increasingly important component of corporate sustainability reporting. Prior research suggests that independent assurance can enhance the credibility, reliability, and perceived usefulness of sustainability disclosures for report users (Hodge et al., 2009; O'Dwyer & Owen, 2005; Simnett et al., 2009). The development of this practice also reflects a broader expansion of assurance services beyond historical financial statements to new forms of corporate reporting (Cohen & Simnett, 2015; O'Dwyer et al., 2011).

Recent professional developments have further formalized this area. ISSA 5000 provides a global framework for sustainability assurance engagements, while related ethical and independence standards clarify the responsibilities of assurance practitioners (International Auditing & Assurance Standards Board, 2024; International Ethics Standards Board for Accountants, 2025). Sustainability reporting standards such as International Financial Reporting Standards (IFRS) S1 and IFRS S2 also increase the importance of professionals who can evaluate sustainability-related information systematically and credibly (IFRS Foundation, 2023a; IFRS Foundation, 2023b). Sustainability assurance therefore involves not only reporting requirements but also the professional capacity needed to assess non-financial information, sustainability-related risks, and supporting evidence.

External auditors possess relevant assurance capabilities, including evidence evaluation, professional judgment, risk assessment, and reporting. However, sustainability assurance also requires familiarity with ESG issues, sustainability reporting frameworks, climate-related disclosures, and forms of evidence that may differ from those commonly encountered in financial statement audits. Auditors may therefore differ in how they evaluate this type of work, how capable they feel of performing it, and whether they intend to participate in such engagements.

This study focuses on behavioral intention because actual sustainability assurance opportunities may not yet be equally available across auditors and audit firms. In such a setting, intention provides a useful indication of an auditor's willingness and motivational readiness to engage in sustainability assurance when opportunities arise. According to the TPB, behavioral intention is closely related to an individual's evaluation of the behavior (Ajzen, 1991; Ajzen, 2020). The present study therefore draws on TPB to examine attitude as an antecedent of intention, but it does not test the full TPB framework.

Subjective norms and perceived behavioral control are not modeled as TPB constructs in this study. Instead, external professional and regulatory influences are examined through perceived regulatory and institutional pressure, while task-specific perceived capability is examined through sustainability assurance self-efficacy. These constructs are not intended as substitutes for subjective norms or perceived behavioral control. Rather, TPB is used specifically to inform the attitude-intention relationship, while institutional theory and self-efficacy theory capture context-specific external signals and task-related perceived capability.

Accordingly, auditors' intention to engage in sustainability assurance is examined as an individual-level outcome shaped by both institutional and individual-level factors. This focus is relevant because the development of credible sustainability assurance depends not only on standards and reporting requirements but also on the availability of professionals who are willing and prepared to perform this work.

2.2 Perceived Regulatory and Institutional Pressure

Institutional theory suggests that organizational and professional behavior is shaped by external pressures arising from regulations, norms, and expectations within the institutional environment (Campbell, 2007; DiMaggio & Powell, 1983). In sustainability reporting and assurance, these pressures may originate from regulators, professional bodies, reporting standards, clients, investors, and other stakeholders. Such signals can increase the perceived legitimacy and professional relevance of independent assurance over sustainability information.

Prior research shows that institutional pressures can shape sustainability-related practices, including environmental management accounting and the voluntary assurance of sustainability reports (Latif et al., 2020; Martínez-Ferrero & García-Sánchez, 2017). More recent studies also show that regulatory and stakeholder pressures are increasingly important in ESG disclosure and assurance environments (Bolognesi et al., 2025; Hummel & Jobst, 2024; Krueger et al., 2024). Evidence from Vietnam similarly indicates that institutional pressures are relevant to sustainability-related organizational practices, although the specific context differs from sustainability assurance (Ngo, 2023). These findings support the relevance of institutional signals in settings where sustainability-related practices are still being formalized.

In this study, perceived regulatory and institutional pressure refers to auditors' perception that regulations, professional standards, regulators, professional bodies, and stakeholder expectations are increasing the need for credible and independently assured sustainability information. This perception may be associated with auditors' attitude toward sustainability assurance. When auditors perceive stronger institutional support and professional expectations for sustainability assurance, they may be more likely to view it as a legitimate, relevant, and valuable area of professional practice.

Perceived regulatory and institutional pressure may also be directly associated with auditors' intention to engage in sustainability assurance. External signals can indicate that sustainability assurance is likely to become more relevant to professional practice and client demand. Such signals may encourage auditors to prepare for, seek exposure to, or participate in sustainability assurance engagements when opportunities arise.

Accordingly, the following hypotheses are proposed:

H1a: Perceived regulatory and institutional pressure is positively associated with auditors' attitude toward sustainability assurance.

H1b: Perceived regulatory and institutional pressure is positively associated with auditors' intention to engage in sustainability assurance.

2.3 Individual Cognitive and Professional Attributes

Beyond institutional pressure, auditors may differ in how they evaluate sustainability assurance because of differences in their perceived knowledge, professional mindset, and perceived capability. These attributes capture distinct aspects of individual readiness. Perceived sustainability literacy reflects how knowledgeable auditors believe they are about sustainability-related issues and reporting frameworks. Professional skepticism reflects a questioning mindset toward claims and evidence. Sustainability assurance self-efficacy reflects auditors' belief in their ability to perform sustainability assurance tasks.

Perceived sustainability literacy refers to auditors' self-assessed knowledge and understanding of sustainability, ESG-related issues, sustainability information, and relevant reporting frameworks. It therefore reflects perceived knowledge rather than objectively demonstrated sustainability knowledge. Prior studies on sustainability literacy show that knowledge and perceived understanding can shape awareness, attitudes, and behavioral responses to sustainability-related issues (Chen et al., 2022; Gericke et al., 2019; Michalos et al., 2012; Ozdemir, 2021). Recent environmental behavior research also suggests that sustainability-related knowledge is associated with more favorable attitudes toward sustainability-oriented actions (Nguyen et al., 2025).

In the audit context, auditors who perceive themselves as more knowledgeable about sustainability and ESG reporting may be better able to recognize the relevance of sustainability disclosures, the credibility risks associated with such information, and the role of independent assurance. Greater perceived familiarity with sustainability reporting frameworks may also help auditors understand how ESG and climate-related information connects with corporate reporting and stakeholder decision-making. These perceptions are expected to contribute to a more favorable evaluation of sustainability assurance as a professional activity.

H2: Perceived sustainability literacy is positively associated with auditors' attitude toward sustainability assurance.

Professional skepticism is a core attribute of audit practice. It involves a questioning mindset, careful evaluation of evidence, and caution before accepting explanations or claims without sufficient support (Hurtt, 2010; Nelson, 2009). Prior auditing research also emphasizes the importance of skepticism in judgment formation and evidence evaluation (Nolder & Kadous, 2018; Quadackers et al., 2014).

Professional skepticism may be particularly relevant to sustainability assurance because sustainability disclosures often contain qualitative information, estimates, forward-looking statements, and claims that can be

difficult to verify. Such features may increase the risk of inconsistency, unsupported assertions, or greenwashing (Hu et al., 2023). Auditors with a stronger skeptical mindset may therefore be more alert to weaknesses in sustainability-related evidence and more likely to recognize the value of independent assurance in improving information credibility.

At the same time, the relationship between skepticism and attitude toward sustainability assurance is not necessarily straightforward. A questioning mindset may also make auditors more sensitive to uncertain metrics, subjective judgments, weak data systems, and difficulties in obtaining sufficient appropriate evidence. Skepticism may therefore increase awareness of both the need for sustainability assurance and the challenges involved in performing it. Nevertheless, because professional skepticism heightens attention to credibility risk and evidence quality, the expected relationship with attitude remains positive.

H3: Professional skepticism is positively associated with auditors' attitude toward sustainability assurance.

Sustainability assurance self-efficacy refers to auditors' perceived capability to perform tasks associated with sustainability assurance. Self-efficacy theory proposes that individuals who believe they can successfully perform a task are more likely to evaluate the task positively and intend to undertake it (Bandura, 1977; Chen et al., 2001). This perspective is particularly relevant to sustainability assurance because auditors may need to apply established assurance skills to unfamiliar reporting frameworks, non-financial evidence, and sustainability-related risks.

Auditors with stronger sustainability assurance self-efficacy may feel more capable of understanding assurance requirements, evaluating sustainability-related evidence, and adapting existing audit skills to sustainability reporting contexts. By reducing uncertainty about whether the work is manageable and professionally feasible, greater perceived capability may contribute to a more favorable attitude toward sustainability assurance. Self-efficacy may also be directly associated with behavioral intention because auditors who believe they can perform sustainability assurance tasks are more likely to be willing to participate when opportunities arise.

H4a: Sustainability assurance self-efficacy is positively associated with auditors' attitude toward sustainability assurance.

H4b: Sustainability assurance self-efficacy is positively associated with auditors' intention to engage in sustainability assurance.

2.4 Attitude Toward Sustainability Assurance and Intention to Engage

Attitude is a central construct in the TPB and refers to an individual's overall evaluation of a behavior as favorable or unfavorable (Ajzen, 1991; Ajzen, 2020). In this study, attitude toward sustainability assurance reflects the extent to which auditors evaluate engagement in sustainability assurance as beneficial, valuable, and professionally worthwhile.

Prior research in accounting and auditing shows that favorable attitudes are associated with stronger behavioral intentions. Studies of auditors' whistleblowing intentions and accountants' adoption of information systems, for example, indicate that individuals are more likely to intend to perform a behavior when they evaluate that behavior positively (Alamin et al., 2020; Alleyne et al., 2013; Tuan Mansor et al., 2021). Although these studies examine different professional behaviors, they support the broader attitude-intention relationship proposed by TPB.

Auditors who view sustainability assurance as a valuable professional activity may be more willing to participate in such engagements when opportunities arise. A favorable attitude may reflect recognition that sustainability assurance can enhance the credibility of sustainability information, support stakeholder confidence, and extend the role of independent assurance to emerging forms of corporate reporting.

This relationship is particularly relevant in a setting where sustainability assurance opportunities are still developing and may not be equally available across auditors. Under such conditions, behavioral intention captures willingness to participate when opportunities become available. The present study therefore draws on the attitude-intention relationship in TPB without treating the model as a full test of the theory.

Accordingly, the following hypothesis is proposed:

H5: Attitude toward sustainability assurance is positively associated with auditors' intention to engage in sustainability assurance.

2.5 Mediating Role of Attitude and Research Model

Attitude may provide a mechanism through which institutional and individual-level factors are associated with auditors' behavioral intention. According to the TPB, individuals are more likely to intend to perform a behavior when they evaluate that behavior favorably (Ajzen, 1991; Ajzen, 2020). In the present context, institutional pressure, perceived sustainability literacy, professional skepticism, and self-efficacy may therefore be associated with intention partly through how auditors evaluate sustainability assurance as a professional activity.

Perceived regulatory and institutional pressure may be indirectly associated with intention through attitude. Stronger regulatory, professional, and stakeholder signals can increase the perceived legitimacy and relevance of sustainability assurance, contributing to a more favorable attitude and, in turn, a stronger intention to participate.

This reasoning is consistent with prior research showing that institutional pressures can shape attitudes and behavioral orientations toward sustainability-related practices (Roxas & Coetzer, 2012; Teo et al., 2003).

Perceived sustainability literacy may also be associated with intention through attitude. Auditors who perceive themselves as more familiar with sustainability concepts, ESG information, and relevant reporting frameworks may better recognize the role of assurance in supporting information credibility, contributing to a more favorable attitude and subsequently a stronger intention to engage.

Professional skepticism may operate through a similar attitudinal mechanism, although the underlying reasoning is more nuanced. A questioning mindset may make auditors more attentive to unsupported sustainability claims, uncertain metrics, and weaknesses in the underlying evidence. This awareness may increase their recognition of the value of independent assurance. To the extent that this recognition contributes to a favorable evaluation of sustainability assurance, professional skepticism may be indirectly associated with intention through attitude.

Sustainability assurance self-efficacy may be associated with intention both directly and indirectly through attitude. Auditors who believe that they can perform sustainability assurance tasks may be more willing to participate in such engagements, while greater perceived capability may also make the work appear more feasible and professionally manageable. A more favorable attitude may therefore provide an additional pathway through which self-efficacy is associated with intention.

The proposed model therefore integrates three complementary theoretical perspectives. Institutional theory explains the role of external regulatory and professional signals, self-efficacy theory explains the role of task-specific perceived capability, and the TPB informs the relationship between attitude and behavioral intention. Perceived sustainability literacy and professional skepticism represent additional individual-level attributes that may shape auditors' evaluation of sustainability assurance. No single perspective captures all of these mechanisms: institutional theory addresses external pressures, self-efficacy theory addresses task-specific perceived capability, and TPB provides the attitudinal pathway to behavioral intention.

Based on these arguments, the following mediation hypotheses are proposed:

H6a: Attitude toward sustainability assurance mediates the relationship between perceived regulatory and institutional pressure and auditors' intention to engage in sustainability assurance.

H6b: Attitude toward sustainability assurance mediates the relationship between perceived sustainability literacy and auditors' intention to engage in sustainability assurance.

H6c: Attitude toward sustainability assurance mediates the relationship between professional skepticism and auditors' intention to engage in sustainability assurance.

H6d: Attitude toward sustainability assurance mediates the relationship between sustainability assurance self-efficacy and auditors' intention to engage in sustainability assurance.

Based on the proposed hypotheses, Figure 1 presents the research model. The model links perceived regulatory and institutional pressure, perceived sustainability literacy, professional skepticism, and sustainability assurance self-efficacy with attitude toward sustainability assurance and intention to engage in sustainability assurance. It includes both direct relationships and indirect relationships through attitude.

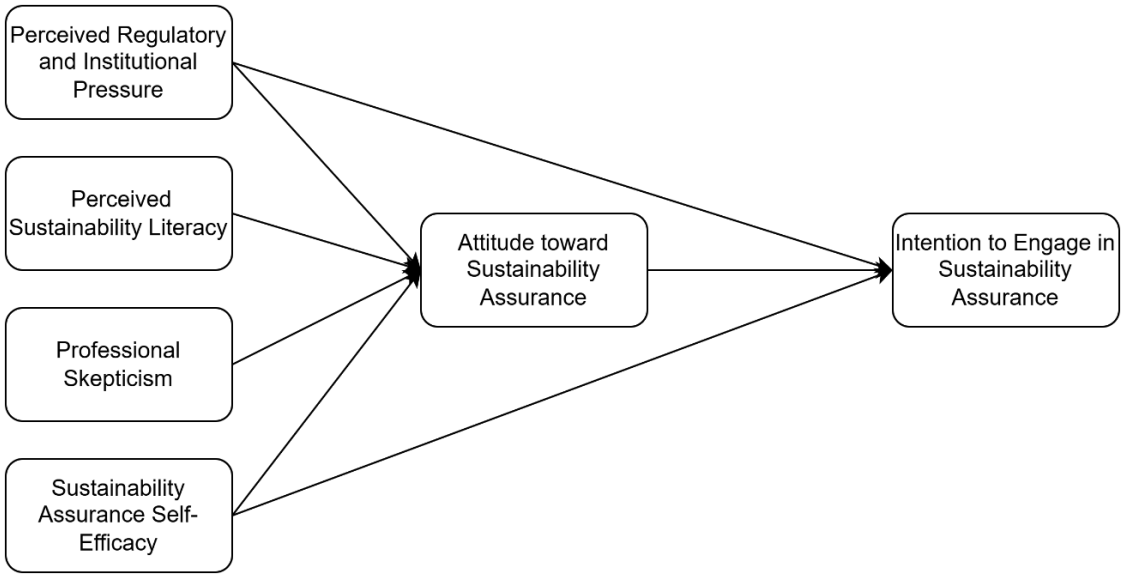


Figure 1. Proposed theoretical framework

3. Methodology

3.1 Research Design

This study used a quantitative, cross-sectional survey design to examine factors associated with external auditors' intention to engage in sustainability assurance. The unit of analysis was the individual external auditor. The study focused on auditors' perceptions, professional attributes, attitudes, and behavioral intentions related to sustainability assurance.

The research model was informed by institutional theory, the TPB, and self-efficacy theory. Institutional theory was used to capture perceived external pressures, while self-efficacy theory addressed auditors' perceived capability to perform sustainability assurance tasks. The TPB informed the relationship between attitude and behavioral intention. The study did not test the full TPB model because subjective norms and perceived behavioral control were not included.

PLS-SEM was used to estimate the proposed relationships among the latent constructs. This approach was appropriate because the model included multiple reflective constructs, direct effects, and mediation effects, and because the analysis focused on explaining variance in attitude toward sustainability assurance and intention to engage in sustainability assurance. Further details on model estimation and assessment are provided in Section 3.5.

3.2 Sample and Data Collection

Data were collected from external auditors working in independent audit firms in Vietnam. Eligible respondents included audit assistants, auditors or senior auditors, audit managers, and partners. The individual external auditor was therefore the unit of analysis. Eligibility was assessed using respondents' self-reported current professional role and employment context. Responses from individuals who did not meet the criterion of working as external auditors in independent audit firms were excluded from the final sample.

A combination of purposive and snowball sampling was used. Purposive sampling was first applied to reach individuals who met the study's eligibility criteria. Initial respondents were contacted through alumni and postgraduate networks, professional contacts, and connections within the professional community of the Vietnam Association of Certified Public Accountants (VACPA). Eligible respondents were then invited to forward the survey link to other external auditors in their professional networks. Snowball sampling was used because a complete sampling frame of external auditors was not available and direct access to this professional population was limited.

The questionnaire was administered online through Google Forms and distributed via Zalo and Gmail between January and March 2026. A total of 600 survey invitations were distributed, and 375 responses were received, yielding a raw response rate of 62.5%. The returned questionnaires were screened for completeness, respondent eligibility, and response validity. Fifty-eight responses were excluded. Of these, 22 contained substantial missing information, 24 were submitted by respondents who did not meet the eligibility criteria, including corporate accountants or internal auditors, and 12 showed internally contradictory or clearly invalid response patterns. The final analytical sample therefore comprised 317 valid responses. This represented 84.5% of the responses received and a usable response rate of 52.8% relative to the 600 invitations distributed.

Table 1 presents the characteristics of the final sample. Auditors or senior auditors represented the largest professional group at 45.7%, followed by audit assistants at 30.0%, audit managers at 17.4%, and partners at 6.9%. In terms of audit experience, 39.4% of respondents had one to three years of experience and 30.0% had four to seven years. Respondents from Non-Big 4 firms accounted for 77.9% of the sample, while 22.1% worked for Big 4 firms. In addition, 36.6% reported having received ESG-related training, whereas 63.4% had not.

Because recruitment partly relied on referrals within professional networks, the snowball procedure may have produced some clustering among respondents with similar professional backgrounds. The sample should therefore not be interpreted as statistically representative of all external auditors in Vietnam. This limitation is considered further in Section 5.4.

3.3 Measurement Development

The measurement instrument was developed by adapting and contextualizing items from prior literature to the sustainability assurance setting. Before responding to the measurement items, participants were provided with a brief definition of sustainability assurance to ensure conceptual clarity. Sustainability assurance was defined as assurance engagements performed on sustainability information, including ESG-related and climate-related disclosures, with the purpose of enhancing the credibility of such information for users.

Because the target respondents were external auditors working in Vietnam, the questionnaire was administered in Vietnamese. Most source items were originally developed in English. A translation and back-translation procedure were therefore conducted following established cross-cultural research practice (Brislin, 1970). The

English questionnaire was first translated into Vietnamese by a bilingual translator familiar with accounting and sustainability terminology. A second independent bilingual translator then translated the Vietnamese version back into English. The original and back-translated versions were compared, and discrepancies in meaning, terminology, or clarity were discussed and resolved before expert review.

Table 1. Respondent profile

Characteristics	Category	Frequency (*n*)	Percentage (%)
Gender	Female	165	52.1
	Male	152	47.9
Position	Audit assistant	95	30.0
	Auditor/Senior auditor	145	45.7
	Audit manager	55	17.4
	Partner	22	6.9
Audit experience	Less than 1 year	43	13.6
	1–3 years	125	39.4
	4–7 years	95	30.0
	8–12 years	39	12.3
	More than 12 years	15	4.7
Firm type	Big 4	70	22.1
	Non-Big 4	247	77.9
ESG training	Yes	116	36.6
	No	201	63.4

Note: ESG = environmental, social, and governance.

Content validity and contextual relevance were assessed through a review by nine experts with knowledge of accounting, auditing, sustainability reporting, and assurance. The initial item pool contained 33 items. Based on the expert review, two semantically redundant items were removed, one item was added to the perceived regulatory and institutional pressure construct to improve its relevance to the Vietnamese regulatory context, and two overlapping professional skepticism items were consolidated into a single item. This process resulted in a provisional 31-item instrument.

The 31-item instrument was then pilot tested with 50 external auditors. The pilot test focused on item clarity, professional terminology, respondent comprehension, and preliminary psychometric performance. Feedback from the pilot participants led to minor refinements in the Vietnamese wording and terminology of several retained items. These revisions were intended to improve clarity and consistency with terminology commonly used in Vietnamese audit practice without changing the underlying conceptual meaning of the constructs.

Preliminary item diagnostics were also examined during the pilot stage, including item loadings, cross-loading patterns, and item-total consistency. Item-retention decisions were based jointly on psychometric performance, conceptual coverage, and contextual relevance rather than on a single statistical criterion. Six items were removed during this stage: one from perceived regulatory and institutional pressure, one from perceived sustainability literacy, two from professional skepticism, one from sustainability assurance self-efficacy, and one from attitude toward sustainability assurance. The final instrument therefore contained 25 items. No further items were removed during the measurement-model assessment of the main sample.

The final questionnaire measured six latent constructs. Perceived regulatory and institutional pressure (RP) was measured using four items. Perceived sustainability literacy (SL) was measured using four items that captured respondents' self-assessed knowledge and familiarity with ESG concepts, sustainability information, and relevant reporting frameworks. This construct therefore reflects perceived rather than objectively demonstrated sustainability literacy. Professional skepticism (PS) was measured using five items that captured a questioning mindset and critical evaluation of sustainability-related claims and evidence. Sustainability assurance self-efficacy (SASE) was measured using four items that reflected auditors' perceived capability to perform sustainability assurance tasks. Attitude toward sustainability assurance (ATSA) was measured using four items that captured respondents' overall evaluation of sustainability assurance as a professional activity. Intention to engage in sustainability assurance (IESA) was measured using four items that reflected willingness, plans, and stated intention to participate in sustainability assurance engagements when opportunities arise.

All items were measured on a five-point Likert scale ranging from 1 = strongly disagree to 5 = strongly agree. The full list of constructs, operational definitions, measurement items, and literature sources is presented in Appendix A.

3.4 Common Method Bias

Because all constructs were measured using self-reported data collected from the same respondents at a single point in time, common method bias was considered a potential concern (Podsakoff et al., 2003; Podsakoff et al.,

2012). Several procedural measures were used to reduce this risk during questionnaire design and administration. Respondents were assured of anonymity and confidentiality and were informed that there were no right or wrong answers. The questionnaire also stated that responses would be used solely for academic research purposes. In addition, the measurement items were worded as clearly and neutrally as possible to reduce ambiguity and evaluation apprehension.

Common method bias was assessed primarily using the full collinearity variance inflation factor (VIF) approach. The full collinearity VIF values ranged from 1.087 to 1.726, all below the conservative threshold of 3.3 (Kock, 2015), providing no indication of problematic common method variance based on this diagnostic. Harman's single-factor test was also conducted as a supplementary assessment. An unrotated principal component analysis of all measurement items produced six components with eigenvalues greater than 1, and the first component accounted for 28.972% of the total variance.

These statistical diagnostics do not rule out common method bias. Given the single-source and cross-sectional nature of the data, some degree of method-related variance may remain. The findings are therefore interpreted cautiously as associations among the study constructs rather than as evidence of causal relationships.

3.5 Data Analysis Method

Data analysis was conducted in two stages. First, SPSS was used for data screening, respondent profile analysis, descriptive statistics, correlation analysis, and the supplementary Harman's single-factor test. These analyses were conducted before estimation of the measurement and structural models.

Second, PLS-SEM was performed using SmartPLS 3. PLS-SEM was selected because the study examined relationships among multiple latent constructs, included both direct and mediation effects, and focused on explaining variance in attitude toward sustainability assurance and intention to engage in sustainability assurance. The method is also suitable for survey data that may not fully satisfy multivariate normality assumptions (Hair et al., 2017). All six latent constructs were specified as reflective measurement models.

The measurement model was assessed before testing the structural relationships. Indicator reliability was evaluated using outer loadings. Internal consistency reliability was assessed using Cronbach's α, Dijkstra–Henseler reliability coefficient (ρA), and composite reliability (ρC). Convergent validity was evaluated using the average variance extracted (AVE), while discriminant validity was assessed using the Fornell-Larcker criterion and the heterotrait-monotrait ratio (HTMT).

After the measurement model was established, the structural model was assessed using inner VIF values, path coefficients, t-values, p-values, 95% confidence intervals (CIs), R^2, and f^2 values. Inner VIF values were examined to assess collinearity among the predictors of each endogenous construct. Statistical significance was evaluated using bootstrapping with 5,000 resamples. The mediating role of attitude toward sustainability assurance was tested using bootstrapped specific indirect effects.

Predictive assessment was conducted in two stages. Stone-Geisser's Q^2 values were first examined to assess predictive relevance within the estimated PLS-SEM model. Because positive Q^2 values alone do not establish out-of-sample predictive performance, PLSpredict was also performed using 10 folds and 10 repetitions (Shmueli et al., 2019). Predictive performance was evaluated using Q^2_predict values and by comparing the root mean squared error (RMSE) of the PLS-SEM predictions with those from a linear-model benchmark.

Two additional robustness analyses were conducted. First, an alternative structural model included direct paths from perceived sustainability literacy and professional skepticism to intention to engage in sustainability assurance. This analysis examined whether their relationships with intention were primarily indirect through attitude or also operated directly. Second, a control-variable model was estimated to assess whether the main structural relationships remained stable after accounting for respondent characteristics. Firm type, ESG-related training, professional position, and audit experience were included as controls for both endogenous constructs. Firm type distinguished Big 4 from Non-Big 4 respondents, ESG-related training distinguished respondents with and without such training, and audit experience distinguished respondents with less than four years of audit experience from those with four or more years.

4. Results

4.1 Descriptive Statistics

Table 2 presents the descriptive statistics and zero-order Pearson correlations among the study constructs. The six constructs are RP, SL, PS, SASE, ATSA, and IESA. Mean scores ranged from 3.499 for SL to 3.839 for PS. All construct means were above the midpoint of the five-point scale. Standard deviations ranged from 0.741 to 0.815, indicating sufficient variation in the responses.

All bivariate correlations were positive. The strongest correlation was between ATSA and IESA ($r = 0.562$, $p < 0.01$), followed by the correlation between SASE and IESA ($r = 0.481$, $p < 0.01$). The remaining correlations

ranged from 0.137 to 0.431. These correlations provide a descriptive overview of the relationships among the constructs; hypothesis testing was conducted using the PLS-SEM structural model reported in the following sections.

Table 2. Descriptive statistics and correlations

Construct	Mean	SD	RP	SL	PS	SASE	ATSA	IESA
RP	3.740	0.784	1					
SL	3.499	0.802	0.232**	1				
PS	3.839	0.741	0.171**	0.216**	1			
SASE	3.737	0.776	0.217**	0.326**	0.168**	1		
ATSA	3.727	0.792	0.431**	0.373**	0.218**	0.413**	1	
IESA	3.664	0.815	0.392**	0.329**	0.137*	0.481**	0.562**	1

Note: $N = 317$. SD = Standard Deviation; RP = perceived regulatory and institutional pressure; SL = perceived sustainability literacy; PS = professional skepticism; SASE = sustainability assurance self-efficacy; ATSA = attitude toward sustainability assurance; IESA = intention to engage in sustainability assurance. Two-tailed significance: ** $p < 0.01$; * $p < 0.05$.

4.2 Measurement Model Assessment

The reflective measurement model was assessed for indicator reliability, internal consistency reliability, convergent validity, and discriminant validity following established PLS-SEM guidelines (Hair et al., 2019; Henseler et al., 2016).

As shown in Table 3, the outer loadings ranged from 0.770 to 0.857, with all values exceeding the recommended threshold of 0.708. Cronbach's α values ranged from 0.816 to 0.865, ρA values from 0.821 to 0.873, and ρC values from 0.878 to 0.903. These results indicate satisfactory internal consistency reliability. The AVE values ranged from 0.643 to 0.695, all above the minimum threshold of 0.50, supporting convergent validity. No additional indicators were removed during the measurement-model assessment of the main sample.

Table 3. Indicator reliability, internal consistency, and convergent validity

Construct	Item	Outer Loading	Cronbach's α	ρA	ρC	AVE
RP	RP1	0.841	0.839	0.840	0.892	0.674
	RP2	0.790				
	RP3	0.809				
	RP4	0.842				
SL	SL1	0.811	0.816	0.821	0.878	0.643
	SL2	0.786				
	SL3	0.785				
	SL4	0.825				
PS	PS1	0.770	0.865	0.873	0.903	0.650
	PS2	0.794				
	PS3	0.857				
	PS4	0.809				
	PS5	0.797				
SASE	SASE1	0.836	0.836	0.838	0.890	0.670
	SASE2	0.836				
	SASE3	0.787				
	SASE4	0.816				
ATSA	ATSA1	0.836	0.853	0.858	0.901	0.695
	ATSA2	0.855				
	ATSA3	0.785				
	ATSA4	0.856				
IESA	IESA1	0.825	0.851	0.851	0.899	0.691
	IESA2	0.837				
	IESA3	0.848				
	IESA4	0.815				

Note: ρA = Dijkstra-Henseler's reliability coefficient; ρC = composite reliability; AVE = average variance extracted; RP = perceived regulatory and institutional pressure; SL = perceived sustainability literacy; PS = professional skepticism; SASE = sustainability assurance self-efficacy; ATSA = attitude toward sustainability assurance; IESA = intention to engage in sustainability assurance.

Discriminant validity was assessed using the Fornell-Larcker criterion and the HTMT. As reported in Table 4, the square root of the AVE for each construct exceeded its correlations with all other constructs, supporting the Fornell-Larcker criterion (Fornell & Larcker, 1981).

Table 4. Fornell-Larcker criterion

Construct	RP	SL	PS	SASE	ATSA	IESA
RP	0.821					
SL	0.235	0.802				
PS	0.172	0.219	0.806			
SASE	0.218	0.332	0.173	0.819		
ATSA	0.433	0.375	0.219	0.417	0.834	
IESA	0.393	0.331	0.137	0.482	0.562	0.831

Note: RP = perceived regulatory and institutional pressure; SL = perceived sustainability literacy; PS = professional skepticism; SASE = sustainability assurance self-efficacy; ATSA = attitude toward sustainability assurance; IESA = intention to engage in sustainability assurance.

In addition, the HTMT values reported in Table 5 ranged from 0.164 to 0.659, all below the conservative threshold of 0.85 (Henseler et al., 2015). Taken together, these results support discriminant validity across the six constructs.

Table 5. HTMT matrix

Construct	RP	SL	PS	SASE	ATSA	IESA
RP	–					
SL	0.280	–				
PS	0.201	0.259	–			
SASE	0.259	0.395	0.201	–		
ATSA	0.509	0.447	0.254	0.490	–	
IESA	0.465	0.394	0.164	0.570	0.659	–

Note: HTMT = heterotrait-monotrait ratio; RP = perceived regulatory and institutional pressure; SL = perceived sustainability literacy; PS = professional skepticism; SASE = sustainability assurance self-efficacy; ATSA = attitude toward sustainability assurance; IESA = intention to engage in sustainability assurance. A dash (–) indicates not applicable.

4.3 Structural Model Assessment

Before testing the structural relationships, collinearity among the predictor constructs was assessed using inner VIF values. The VIF values ranged from 1.077 to 1.422, indicating no problematic collinearity among the predictors of ATSA and IESA.

The structural relationships were then assessed using bootstrapping with 5,000 resamples. As reported in Table 6, RP was positively associated with ATSA ($\beta = 0.316$, $t = 6.357$, $p < 0.001$) and IESA ($\beta = 0.171$, $t = 3.456$, $p = 0.001$), supporting H1a and H1b. SL was positively associated with ATSA ($\beta = 0.195$, $t = 3.716$, $p < 0.001$), supporting H2.

PS was not significantly associated with ATSA ($\beta = 0.075$, $t = 1.537$, $p = 0.124$), and H3 was therefore not supported. SASE was positively associated with both ATSA ($\beta = 0.270$, $t = 5.104$, $p < 0.001$) and IESA ($\beta = 0.292$, $t = 6.037$, $p < 0.001$), supporting H4a and H4b. ATSA was also positively associated with IESA ($\beta = 0.366$, $t = 6.840$, $p < 0.001$), supporting H5.

Table 6. Structural model and hypothesis testing

Hypothesis	Structural Path	VIF	*β*	*t*-Value	*p*-Value	95% CI (Lower)	95% CI (Upper)	Result
H1a	RP → ATSA	1.097	0.316	6.357	<0.001	0.214	0.409	Supported
H1b	RP → IESA	1.234	0.171	3.456	0.001	0.072	0.266	Supported
H2	SL → ATSA	1.188	0.195	3.716	<0.001	0.083	0.290	Supported
H3	PS → ATSA	1.077	0.075	1.537	0.124	-0.022	0.171	Not supported
H4a	SASE → ATSA	1.160	0.270	5.104	<0.001	0.167	0.374	Supported
H4b	SASE → IESA	1.212	0.292	6.037	<0.001	0.199	0.387	Supported
H5	ATSA → IESA	1.422	0.366	6.840	<0.001	0.258	0.466	Supported

Note: VIF, variance inflation factor, it refers to structural-model collinearity among the predictors of each endogenous construct. CI = confidence interval; RP = perceived regulatory and institutional pressure; SL = perceived sustainability literacy; PS = professional skepticism; SASE = sustainability assurance self-efficacy; ATSA = attitude toward sustainability assurance; IESA = intention to engage in sustainability assurance.

Figure 2 presents the standardized structural path coefficients, indicator loadings, and R^2 values. The model explained 33.9% of the variance in ATSA and 41.4% of the variance in IESA. These results indicate that the proposed predictors accounted for a meaningful proportion of variation in both endogenous constructs, while leaving additional variance to be explained by factors outside the model.

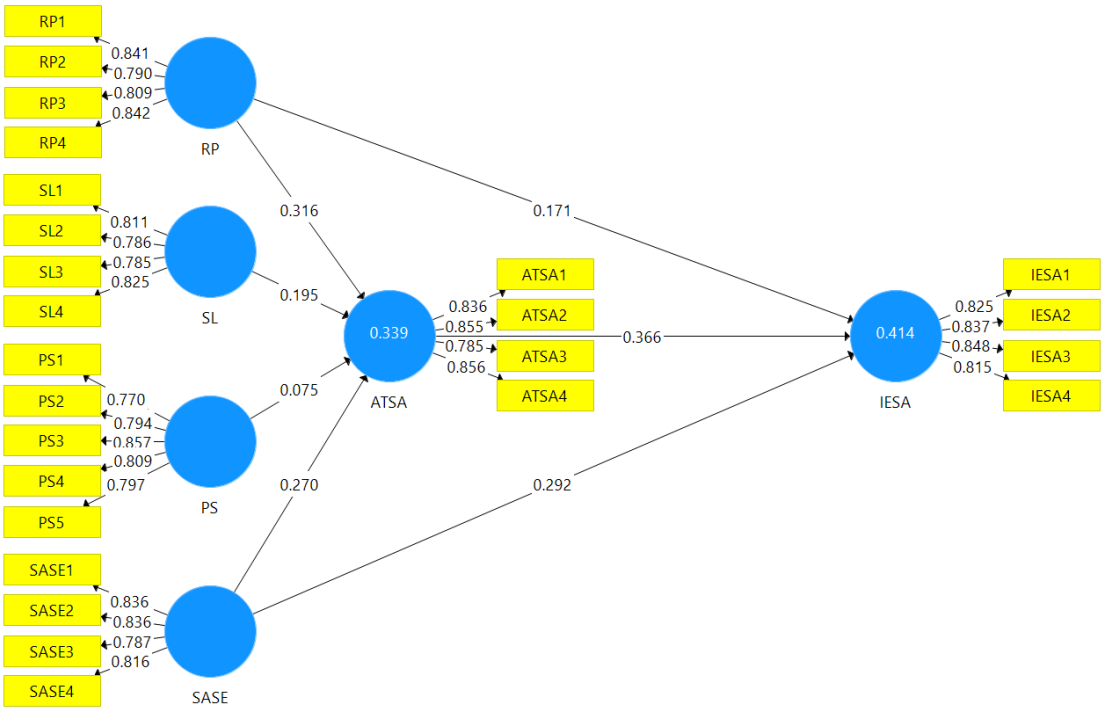


Figure 2. Structural model results

Note: RP = perceived regulatory and institutional pressure; SL = perceived sustainability literacy; PS = professional skepticism; SASE = sustainability assurance self-efficacy; ATSA = attitude toward sustainability assurance; IESA = intention to engage in sustainability assurance.

Effect sizes were examined using f^2 values. As shown in Table 7, ATSA had the largest effect size for IESA (f^2 = 0.161). RP showed an f^2 value of 0.138 for ATSA, while SASE showed f^2 values of 0.095 for ATSA and 0.120 for IESA. The f^2 values for SL and RP in their remaining relationships were smaller, and the value for PS → ATSA was 0.008, consistent with the non-significant structural path.

Table 7. Effect size (f^2) for each path

Exogenous Construct	Endogenous Construct	f^2 Value
RP	ATSA	0.138
RP	IESA	0.040
SL	ATSA	0.048
PS	ATSA	0.008
SASE	ATSA	0.095
SASE	IESA	0.120
ATSA	IESA	0.161

Note: RP = perceived regulatory and institutional pressure; SL = perceived sustainability literacy; PS = professional skepticism; SASE = sustainability assurance self-efficacy; ATSA = attitude toward sustainability assurance; IESA = intention to engage in sustainability assurance.

4.4 Mediation Analysis

The mediating role of attitude toward sustainability assurance was examined using bootstrapped specific indirect effects. As reported in Table 8, RP had a positive indirect association with IESA through ATSA (β = 0.116, t = 4.724, $p < 0.001$, 95% CI [0.074, 0.169]), supporting H6a. SL also had a significant positive indirect association with IESA through ATSA (β = 0.071, t = 3.070, p = 0.002, 95% CI [0.029, 0.119]), supporting H6b.

In contrast, the indirect association between PS and IESA through ATSA was not statistically significant (β = 0.028, t = 1.497, p = 0.134, 95% CI [−0.007, 0.066]); therefore, H6c was not supported. SASE showed a significant positive indirect association with IESA through ATSA (β = 0.099, t = 4.186, $p < 0.001$, 95% CI [0.059, 0.153]), supporting H6d.

Overall, the results support an indirect role of ATSA in the relationships of RP, SL, and SASE with IESA, whereas no corresponding indirect relationship was found for PS.

Table 8. Mediation analysis results

Hypothesis	Indirect Path	*β*	*t*-Value	*p*-Value	95% CI (Lower)	95% CI (Upper)	Result
H6a	RP → ATSA → IESA	0.116	4.724	<0.001	0.074	0.169	Supported
H6b	SL → ATSA → IESA	0.071	3.070	0.002	0.029	0.119	Supported
H6c	PS → ATSA → IESA	0.028	1.497	0.134	−0.007	0.066	Not supported
H6d	SASE → ATSA → IESA	0.099	4.186	<0.001	0.059	0.153	Supported

Note: CI = confidence interval; RP = perceived regulatory and institutional pressure; SL = perceived sustainability literacy; PS = professional skepticism; SASE = sustainability assurance self-efficacy; ATSA = attitude toward sustainability assurance; IESA = intention to engage in sustainability assurance.

4.5 Predictive Assessment

Predictive assessment was conducted using both Stone-Geisser's Q^2 and PLSpredict. The Stone-Geisser Q^2 values were positive for both endogenous constructs, with $Q^2 = 0.228$ for ATSA and $Q^2 = 0.281$ for IESA. These results indicate predictive relevance within the estimated PLS-SEM model.

PLSpredict was then used to assess out-of-sample predictive performance. As reported in Table 9, all Q^2_predict values were greater than zero, ranging from 0.164 to 0.265 across the eight indicators of ATSA and IESA. In addition, the PLS-SEM RMSE values were lower than those of the linear-model benchmark for all eight indicators. Taken together, these results provide consistent evidence that the model has out-of-sample predictive capability for both endogenous constructs.

Table 9. PLSpredict assessment

Indicator	Q^2_Predict	PLS-SEM RMSE	LM RMSE
ATSA1	0.226	0.824	0.846
ATSA2	0.214	0.859	0.895
ATSA3	0.164	0.856	0.880
ATSA4	0.265	0.829	0.864
IESA1	0.212	0.903	0.947
IESA2	0.235	0.839	0.852
IESA3	0.224	0.828	0.850
IESA4	0.213	0.898	0.921

Note: PLS-SEM = partial least squares structural equation modeling; RMSE = root mean squared error; LM = linear-model benchmark; ATSA = attitude toward sustainability assurance; IESA = intention to engage in sustainability assurance. Q^2_predict values greater than zero indicate that the PLS-SEM model provides predictive information for the corresponding indicator.

4.6 Robustness Analyses

4.6.1 Alternative direct-path model

To assess whether SL and PS were directly associated with IESA, an alternative structural model was estimated by adding direct paths from SL and PS to IESA.

As shown in Table 10, the main structural relationships remained consistent with those in the baseline model. The additional direct path from SL to IESA was not statistically significant ($\beta = 0.076$, $t = 1.500$, $p = 0.134$, 95% CI [−0.028, 0.173]). The direct path from PS to IESA was also not significant ($\beta = -0.032$, $t = 0.661$, $p = 0.509$, 95% CI [−0.131, 0.057]).

Table 10. Robustness analysis: Alternative direct-path model

Structural Path	*β*	*t*-Value	*p*-Value	95% CI Lower	95% CI Upper	Result
RP → ATSA	0.316	6.325	<0.001	0.211	0.411	Significant
RP → IESA	0.168	3.382	0.001	0.065	0.261	Significant
SL → ATSA	0.195	3.717	<0.001	0.085	0.291	Significant
SL → IESA	0.076	1.500	0.134	−0.028	0.173	Not significant
PS → ATSA	0.074	1.499	0.134	−0.025	0.169	Not significant
PS → IESA	−0.032	0.661	0.509	−0.131	0.057	Not significant
SASE → ATSA	0.270	5.192	<0.001	0.163	0.367	Significant
SASE → IESA	0.279	5.489	<0.001	0.180	0.380	Significant
ATSA → IESA	0.352	6.577	<0.001	0.246	0.454	Significant

Note: CI = confidence interval; RP = perceived regulatory and institutional pressure; SL = perceived sustainability literacy; PS = professional skepticism; SASE = sustainability assurance self-efficacy; ATSA = attitude toward sustainability assurance; IESA = intention to engage in sustainability assurance.

When considered together with the mediation results reported in Section 4.4, these findings are consistent with SL being associated with IESA primarily through ATSA. In contrast, PS showed neither a significant direct association with IESA nor a significant indirect association through ATSA.

4.6.2 Control-variable robustness analysis

A second robustness analysis examined whether the main structural relationships remained stable after accounting for respondent characteristics. Firm type, ESG-related training, professional position, and audit experience were included as controls for both ATSA and IESA.

As reported in Table 11, the substantive conclusions of the baseline model remained unchanged after the controls were introduced. RP remained positively associated with ATSA ($\beta = 0.315$, $p < 0.001$) and IESA ($\beta = 0.173$, $p < 0.001$). SL remained positively associated with ATSA ($\beta = 0.196$, $p < 0.001$), while PS remained non-significant ($\beta = 0.061$, $p = 0.214$). SASE remained positively associated with both ATSA ($\beta = 0.278$, $p < 0.001$) and IESA ($\beta = 0.295$, $p < 0.001$), and ATSA remained positively associated with IESA ($\beta = 0.367$, $p < 0.001$).

Among the control variables, only professional position was significantly associated with ATSA ($\beta = 0.187$, $p = 0.001$). The remaining control paths were not statistically significant at the 5% level. The inclusion of the controls increased R^2 from 0.339 to 0.368 for ATSA and from 0.414 to 0.422 for IESA. Overall, the main structural findings remained stable after accounting for these respondent characteristics.

Table 11. Robustness analysis with respondent-level controls

Structural Path	*β*	*t*-Value	*p*-Value	95% CI Lower	95% CI Upper
RP → ATSA	0.315	6.093	<0.001	0.212	0.415
RP → IESA	0.173	3.482	<0.001	0.075	0.270
SL → ATSA	0.196	3.734	<0.001	0.094	0.299
PS → ATSA	0.061	1.242	0.214	−0.027	0.167
SASE → ATSA	0.278	5.277	<0.001	0.173	0.380
SASE → IESA	0.295	5.970	<0.001	0.200	0.393
ATSA → IESA	0.367	6.719	<0.001	0.260	0.471
Big 4 → ATSA	−0.020	0.417	0.677	−0.110	0.076
Big 4 → IESA	0.065	1.442	0.149	−0.025	0.153
ESG training → ATSA	0.023	0.382	0.702	−0.095	0.137
ESG training → IESA	−0.106	1.934	0.053	−0.214	0.003
Audit experience → ATSA	−0.039	0.542	0.588	−0.178	0.099
Audit experience → IESA	0.035	0.548	0.583	−0.091	0.159
Professional position → ATSA	0.187	3.208	0.001	0.076	0.303
Professional position → IESA	0.009	0.165	0.869	−0.094	0.117

Note: CI = confidence interval; RP = perceived regulatory and institutional pressure; SL = perceived sustainability literacy; PS = professional skepticism; SASE = sustainability assurance self-efficacy; ATSA = attitude toward sustainability assurance; IESA = intention to engage in sustainability assurance; ESG = environmental, social, and governance. Audit experience was coded as less than four years versus four or more years. Firm type, ESG-related training, professional position, and audit experience were included as controls for both endogenous constructs. Professional position was included as an ordinal control variable coded as 1 = audit assistant, 2 = auditor/senior auditor, 3 = audit manager, and 4 = partner.

5. Discussion

5.1 Discussion of Findings

This study examined how institutional and individual-level factors are associated with external auditors' readiness to engage in sustainability assurance in Vietnam. The findings indicate that auditors' willingness to engage is associated with both the external environment in which they operate and how they evaluate their own capability and the professional value of sustainability assurance.

Perceived regulatory and institutional pressure was positively associated with both attitude toward sustainability assurance and intention to engage. This finding is consistent with institutional theory, which emphasizes the role of regulatory, normative, and stakeholder expectations in shaping professional behavior (Campbell, 2007; DiMaggio & Powell, 1983). This interpretation is particularly relevant in Vietnam, where sustainability reporting requirements are expanding while formal sustainability assurance remains uneven. Under such conditions, auditors may use regulatory developments, professional standards, and stakeholder expectations as signals that sustainability assurance is becoming a more established part of corporate reporting and accountability. The finding therefore has implications beyond the audit profession because the availability of auditors willing to undertake this work can influence the capacity of the reporting system to provide more credible sustainability information.

Perceived sustainability literacy was positively associated with attitude but did not show a significant direct association with intention in the alternative direct-path model. This pattern suggests that self-assessed familiarity with sustainability concepts, ESG information, and reporting frameworks is more closely related to how auditors

evaluate sustainability assurance than to whether they immediately intend to participate in such engagements. Auditors who perceive themselves as more familiar with sustainability-related information may be better able to understand why independent assurance matters for reporting credibility and stakeholder use of ESG information. This interpretation is consistent with research linking sustainability-related knowledge and awareness with more favorable sustainability attitudes (Chen et al., 2022; Gericke et al., 2019; Michalos et al., 2012). Importantly, however, the construct captures perceived rather than objectively demonstrated sustainability literacy. The findings should therefore not be interpreted as evidence that higher actual sustainability competence necessarily leads to stronger assurance intention.

The non-significant result for professional skepticism is particularly informative. Professional skepticism is central to audit judgment because it promotes critical evaluation of claims and evidence (Nelson, 2009; Nolder & Kadous, 2018). However, recognizing weaknesses in sustainability information is conceptually different from evaluating sustainability assurance as an attractive or feasible professional activity. Skeptical auditors may be more alert to unsupported ESG claims, uncertain metrics, qualitative evidence, forward-looking disclosures, and weaknesses in sustainability data systems. These same features may also make sustainability assurance appear difficult to perform. Professional skepticism may therefore increase awareness of assurance risk without necessarily producing a more favorable attitude toward undertaking this type of engagement.

This interpretation should also be considered in light of the study design. The survey did not collect information on respondents' prior direct experience with sustainability assurance engagements, and several skepticism items were framed in sustainability-related contexts. Respondents without direct experience may therefore have evaluated these situations on a prospective or hypothetical basis. This cannot be established from the present data, but it provides one plausible explanation for why professional skepticism was not significantly associated with attitude or indirectly with intention. The result suggests that a skeptical mindset should not be assumed to translate automatically into readiness to enter a developing assurance domain.

Sustainability assurance self-efficacy showed a different pattern. It was positively associated with both attitude and intention, and it also had an indirect association with intention through attitude. This finding is consistent with self-efficacy theory, which emphasizes the importance of perceived capability in approaching unfamiliar or demanding tasks (Bandura, 1977; Chen et al., 2001). Auditors who believe that they can manage sustainability-related reporting frameworks, non-financial evidence, and measurement demands may be more likely to view sustainability assurance as professionally feasible and to express willingness to participate. The result therefore highlights perceived capability as an important element of professional readiness, although it should not be interpreted as evidence of objectively demonstrated competence.

Attitude toward sustainability assurance was positively associated with intention, consistent with the attitude-intention relationship in the TPB (Ajzen, 1991; Ajzen, 2020). The present study uses this specific relationship without testing the full TPB framework. The mediation results reinforce this interpretation: attitude provided an indirect pathway linking perceived regulatory and institutional pressure, perceived sustainability literacy, and sustainability assurance self-efficacy with intention. For perceived sustainability literacy, the absence of a significant direct path to intention in the alternative model further suggests that its relationship with intention operates primarily through auditors' evaluation of sustainability assurance. By contrast, no corresponding indirect relationship was found for professional skepticism.

The control-variable analysis provides additional support for the stability of these findings. The main structural relationships remained substantively unchanged after accounting for firm type, ESG-related training, professional position, and audit experience. Among these controls, only professional position was significantly associated with attitude toward sustainability assurance, while the remaining control paths were not statistically significant at the 5% level. These results suggest that the principal relationships observed in the model are not solely attributable to the uneven composition of the sample across Big 4 and Non-Big 4 firms, ESG training backgrounds, or audit experience groups. Nevertheless, the non-probability sampling design means that the findings should not be generalized to the entire population of Vietnamese external auditors without caution.

Taken together, the findings suggest that sustainability assurance capacity depends not only on formal standards and reporting requirements but also on how auditors perceive the relevance, feasibility, and professional value of this work. Stronger institutional signals, perceived task capability, and favorable attitudes are associated with greater readiness to engage, while perceived sustainability literacy contributes mainly through attitude and professional skepticism alone does not appear sufficient to generate such readiness.

5.2 Theoretical Implications

This study offers several theoretical implications for sustainability assurance research. First, it extends the literature by shifting attention from organizational assurance decisions to the individual-level professional capacity underlying sustainability assurance. Prior studies have largely examined assurance demand, provider choice, assurance quality, and the credibility effects of assurance (Bentley-Goode et al., 2025; Cohen & Simnett, 2015; Gipper et al., 2025; Simnett et al., 2009). The present study complements this literature by showing that auditors'

readiness to engage in sustainability assurance is associated with how they perceive institutional signals, their perceived sustainability-related knowledge and task-specific capability, and the professional value of assurance work. This individual-level perspective is theoretically relevant because the development of credible sustainability assurance depends not only on reporting requirements and organizational demand but also on the professionals available to perform assurance engagements.

Second, the study contributes an integrated explanation of auditor readiness by combining institutional, cognitive, professional, and behavioral mechanisms within a single framework. Institutional theory captures external regulatory and professional signals, self-efficacy theory addresses task-specific perceived capability, and the TPB informs the attitude-intention pathway, while perceived sustainability literacy and professional skepticism represent additional individual-level attributes. Their combination provides a more differentiated account of how external conditions and individual perceptions are associated with auditors' readiness to participate in sustainability assurance.

Third, the findings refine the application of the TPB to sustainability assurance. This study does not test the full TPB framework and does not treat perceived regulatory and institutional pressure or sustainability assurance self-efficacy as substitutes for subjective norms or perceived behavioral control. Rather, TPB is used specifically to explain the attitude-intention relationship. The positive association between attitude and intention, together with the significant indirect relationships through attitude, indicates that auditors' evaluation of sustainability assurance is an important mechanism linking several antecedents with behavioral intention, even when the broader TPB structure is not modeled.

Fourth, the findings distinguish perceived sustainability literacy from sustainability assurance self-efficacy. Perceived sustainability literacy was associated with attitude and showed a significant indirect association with intention through attitude, while its additional direct path to intention was not statistically significant. By contrast, sustainability assurance self-efficacy was associated with both attitude and intention. This distinction suggests that perceiving oneself as knowledgeable about sustainability-related concepts is not theoretically equivalent to believing that one can perform sustainability assurance tasks. Sustainability-related understanding may shape how auditors evaluate the relevance and value of assurance, whereas task-specific self-efficacy is more directly connected to perceived feasibility and willingness to participate. This distinction is especially important because perceived sustainability literacy in this study reflects self-assessed knowledge rather than objectively demonstrated competence.

Finally, the non-significant findings for professional skepticism provide a theoretical boundary to the role of traditional audit attributes in emerging sustainability assurance settings. Professional skepticism remains important for evidence evaluation and audit judgment, but a questioning mindset does not necessarily translate into a favorable attitude toward entering a new assurance domain. Skepticism may heighten awareness of uncertain metrics, weak evidence, forward-looking disclosures, and sustainability data limitations without increasing the perceived feasibility or attractiveness of sustainability assurance work. The findings therefore suggest that theoretical models of sustainability assurance readiness should distinguish between attributes that support the execution of assurance tasks and factors that shape willingness to engage in such work.

5.3 Practical Implications

The findings have practical implications for regulators, professional bodies, audit firms, and educators involved in the development of sustainability assurance. The results most directly point to three areas: the clarity of institutional signals surrounding sustainability assurance, auditors' perceived capability to perform sustainability assurance tasks, and auditors' evaluation of the professional relevance of this work. These implications concern areas that may warrant attention rather than interventions shown by the present study to improve actual professional competence or subsequent participation.

First, the positive associations of perceived regulatory and institutional pressure with both attitude and intention suggest that clear and consistent regulatory and professional signals may be important as sustainability assurance develops. In settings such as Vietnam, where sustainability reporting requirements are expanding while assurance practices remain uneven, regulators and professional bodies could provide implementation-oriented guidance to clarify professional expectations and the application of emerging international standards. Possible measures include Vietnamese-language guidance, technical explanations of ISSA 5000, and coordinated communication among relevant regulatory and professional institutions.

Second, the positive associations of sustainability assurance self-efficacy with both attitude and intention indicate that auditors' perceived task capability deserves attention. Audit firms may therefore consider professional development and supervised exposure that help practitioners become more familiar with sustainability-related evidence, reporting frameworks, and assurance requirements. Perceived sustainability literacy showed a different pattern: it was associated with attitude but not directly with intention in the robustness analysis. Universities, professional associations, and audit firms may therefore also consider strengthening exposure to sustainability reporting, ESG information, and assurance concepts within accounting and auditing education. Recent professional

initiatives similarly emphasize sustainability-related competencies in accounting education and professional development (International Federation of Accountants, 2024; International Federation of Accountants, 2025), while calls to integrate ESG and sustainability reporting into accounting education have also emerged in Vietnam (Vu, 2025). Importantly, both constructs are self-assessed; the findings do not demonstrate that these forms of professional development increase actual competence or behavioral intention.

The professional skepticism finding provides a more limited implication. Its non-significant associations with attitude and indirect intention suggest that skepticism alone should not be treated as sufficient evidence of readiness to engage in sustainability assurance. Professional development may need to connect established audit judgment skills with the specific characteristics of sustainability information, such as qualitative evidence, uncertain metrics, forward-looking disclosures, and data-quality limitations. Similarly, the positive association between attitude and intention indicates that perceived professional relevance matters alongside perceived capability. Audit firms and professional bodies may therefore help practitioners understand how sustainability assurance relates to the credibility of ESG information and the broader development of sustainability reporting, without assuming that communication or training alone will increase engagement.

Overall, the findings indicate that professional readiness is associated with institutional clarity, perceived task capability, sustainability-related understanding, and favorable evaluation of sustainability assurance. These areas provide a basis for considering future capacity-building efforts, while the effectiveness of specific interventions and their effects on demonstrated competence or actual participation require separate evaluation.

5.4 Limitations and Future Research

This study has several limitations that should be considered when interpreting the findings. First, the data were collected using a cross-sectional survey design. The observed relationships therefore represent associations among the study constructs and should not be interpreted as evidence of causal effects. Future research could use longitudinal or repeated-measures designs to examine whether auditors' attitudes and intentions change as sustainability reporting requirements and assurance practices become more institutionalized.

Second, the study relied on self-reported data collected from the same respondents. Although procedural measures and statistical diagnostics were used to address potential common method bias, some method-related variance may remain. This limitation is particularly relevant to perceived sustainability literacy and sustainability assurance self-efficacy, both of which capture respondents' own assessments rather than objectively demonstrated knowledge or capability. Future studies could combine self-reported measures with objective sustainability knowledge tests, task-based assessments, experiments, interviews, or other behavioral measures to distinguish perceived readiness from demonstrated competence more clearly.

Third, the study did not collect information on respondents' prior direct experience with sustainability assurance engagements. This limitation is particularly relevant to the interpretation of professional skepticism. Because several skepticism items were framed in sustainability-specific contexts, respondents without direct sustainability assurance experience may have evaluated these situations prospectively or hypothetically rather than on the basis of actual engagement experience. The non-significant relationships involving professional skepticism should therefore be interpreted with this limitation in mind. Future studies could distinguish auditors with and without prior sustainability assurance experience and examine whether experience moderates the relationships among skepticism, attitude, self-efficacy, and intention.

Fourth, purposive and snowball sampling were used to recruit external auditors working in independent audit firms in Vietnam. Although this approach facilitated access to a specialized professional population for which a complete sampling frame was not available, it limits statistical generalizability. Snowball recruitment may also have produced some clustering among respondents from similar professional networks or organizational backgrounds. The control-variable analysis indicated that the main structural relationships remained broadly stable after accounting for firm type, ESG-related training, professional position, and audit experience, but this analysis does not eliminate the limitations associated with non-probability sampling. Future research could use broader sampling frames, professional association databases, stratified sampling, or direct collaboration with audit firms to improve sample representativeness.

Fifth, the study focused on external auditors in Vietnam. The Vietnamese setting is informative because sustainability reporting is expanding while formal sustainability assurance remains uneven, but the findings may not generalize to jurisdictions with different regulatory systems, assurance requirements, audit market structures, or levels of sustainability reporting maturity. Comparative research across countries could examine whether the relative importance of institutional pressure, perceived sustainability literacy, self-efficacy, and attitude differs between jurisdictions with mandatory assurance requirements and those where sustainability assurance remains voluntary or developing.

Sixth, the study examined intention to engage in sustainability assurance rather than actual professional behavior. Behavioral intention is relevant where engagement opportunities are not yet equally available, but intention does not necessarily translate into subsequent participation. Future research could follow auditors over time or collect

information on actual sustainability assurance assignments to examine when and under what organizational conditions intention leads to professional involvement.

Finally, the proposed model was deliberately selective and did not test the full TPB. Subjective norms and perceived behavioral control were not included as TPB constructs, and other potentially relevant factors such as organizational support, client demand, ethical climate, career expectations, and firm-level assurance capability were also outside the scope of the study. Future research could examine these factors alongside direct sustainability assurance experience and objective sustainability knowledge. Such extensions would help determine whether the relationships identified in this study remain stable when broader individual, organizational, and institutional conditions are considered.

6. Conclusions

This study examined external auditors' readiness to engage in sustainability assurance in Vietnam using a model informed by institutional theory, self-efficacy theory, and the attitude-intention relationship in the TPB. Using survey data from 317 external auditors, the study examined how institutional and individual-level factors are associated with readiness to engage in sustainability assurance.

The findings show that perceived regulatory and institutional pressure and sustainability assurance self-efficacy were positively associated with both attitude and intention. Perceived sustainability literacy was positively associated with attitude and showed a significant indirect association with intention through attitude, while its additional direct path to intention was not statistically significant. Attitude was positively associated with intention and provided an indirect pathway linking several antecedents with behavioral intention. Professional skepticism, by contrast, was not significantly associated with attitude or indirectly with intention.

Overall, the findings indicate that sustainability assurance capacity depends not only on reporting standards and demand for assurance services but also on auditors' perceptions of institutional signals, task-specific capability, and the professional relevance of sustainability assurance. In an emerging reporting environment such as Vietnam, these individual-level factors are relevant to the professional capacity needed to support credible sustainability reporting and independent assurance. The findings should nevertheless be interpreted as associations based on self-reported perceptions and behavioral intention rather than as evidence of demonstrated professional competence or actual participation in sustainability assurance engagements.

Informed Consent Statement

Informed consent was obtained from all subjects involved in the study.

Ethical Approval

The study was conducted in accordance with the academic integrity regulations of Thuongmai University, established under Decision No. 828/QĐ-ĐHTM dated 16 May 2024. Under the University's applicable institutional requirements, formal ethics committee approval was not required for this non-interventional online survey of adult external auditors. Participation was voluntary, informed consent was obtained from all respondents, and their responses were treated confidentially and used solely for academic research purposes.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this work, the author used ChatGPT in order to enhance language fluency and readability. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Appendix

Appendix A. Measurement instrument

Definition provided to respondents: In this survey, sustainability assurance refers to assurance engagements performed on sustainability information, including ESG-related and climate-related disclosures, with the purpose of enhancing the credibility of such information for users.

Response scale: All items were measured on a five-point Likert scale, where 1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, and 5 = strongly agree.

Measurement note: The questionnaire measures respondents' perceptions, self-assessed knowledge, professional mindset, attitude, and behavioral intention. Items were either adapted from established measurement guidance/scales or developed and contextualized from prior theoretical and empirical literature. Where an item was not taken verbatim from a single scale, the source column states the theoretical and contextual basis used for item development.

· **Perceived regulatory and institutional pressure**

Operational definition: Auditors' perception that regulations, professional standards, regulators, professional bodies, and stakeholders are increasing the need for credible and independently assured sustainability information.

Measurement basis: Developed based on institutional theory and institutional pressure literature, and contextualized to sustainability reporting and assurance developments (Campbell, 2007; DiMaggio & Powell, 1983; Gipper et al., 2025; Hummel & Jobst, 2024; International Auditing & Assurance Standards Board, 2024; Krueger et al., 2024; Latif et al., 2020; Martínez-Ferrero & García-Sánchez, 2017).

Table A1. Measurement items for RP

Code	Measurement Item	Item-Level Adaptation/Source Note
RP1	I perceive that sustainability reporting regulations are increasing the need for independent assurance on sustainability information.	Regulatory/coercive pressure item developed from institutional pressure literature and contextualized to sustainability reporting regulation and assurance demand.
RP2	I perceive that professional standards are increasing expectations for auditors to develop sustainability assurance competencies.	Professional/normative pressure item contextualized with sustainability assurance standards and professional competency expectations.
RP3	I perceive that regulators and professional bodies increasingly emphasize the credibility of sustainability disclosures.	Regulator/professional-body pressure item developed from institutional and sustainability reporting regulation literature.
RP4	I perceive that stakeholder expectations for credible sustainability information are increasing the demand for independent assurance.	Stakeholder expectation item developed from sustainability assurance demand and institutional pressure literature.

Note: RP = perceived regulatory and institutional pressure.

· **Perceived sustainability literacy**

Operational definition: Auditors' self-assessed knowledge and understanding of sustainability, ESG-related issues, sustainability information, and sustainability reporting frameworks relevant to assurance engagements.

Measurement basis: Developed and contextualized from sustainability literacy and sustainability education literature (Chen et al., 2022; Gericke et al., 2019; Michalos et al., 2012; Ozdemir, 2021). The items measure perceived/self-assessed literacy, not objective knowledge test performance. The reporting-framework familiarity item was further contextualized with reference to IFRS S1 and IFRS S2 (IFRS Foundation, 2023a; IFRS Foundation, 2023b).

Table A2. Measurement items for SL

Code	Measurement Item	Item-Level Adaptation/Source Note
SL1	I have adequate knowledge of sustainability and ESG-related concepts.	Self-assessed conceptual knowledge item based on sustainability literacy literature.
SL2	I understand the types of sustainability information commonly disclosed by organizations.	Self-assessed understanding of sustainability disclosure content; contextualized to corporate reporting.
SL3	I understand how environmental, social, and governance (ESG) issues are connected to corporate reporting.	Self-assessed integrative understanding of ESG issues and corporate reporting.
SL4	I am familiar with key sustainability reporting frameworks or standards relevant to assurance engagements.	Self-assessed familiarity with reporting frameworks/standards; contextualized to International Financial Reporting Standards (IFRS) S1, IFRS S2, International Standard on Sustainability Assurance 5000 (ISSA 5000), and sustainability education guidance.

Note: SL = perceived sustainability literacy.

· **Professional skepticism in sustainability assurance**

Operational definition: Auditors' questioning mindset and tendency to critically evaluate evidence, explanations, and sustainability-related claims before reaching assurance conclusions.

Measurement basis: Adapted from professional skepticism and audit judgment literature, particularly Hurtt's professional skepticism scale and related audit skepticism studies, with contextualization to sustainability disclosures, ESG claims, and assurance evidence (Hurtt, 2010; Nelson, 2009; Nolder & Kadous, 2018; Quadackers et al., 2014).

Table A3. Measurement items for PS in sustainability assurance

Code	Measurement Item	Item-Level Adaptation/Source Note
PS1	When evaluating assurance evidence, I withhold conclusions until sufficient evidence has been obtained.	Adapted from general professional skepticism/evidence evaluation logic.
PS2	I require convincing evidence before accepting sustainability-related claims.	Contextualized skepticism item focused on sustainability-related claims and greenwashing risk.
PS3	I question explanations provided by clients or management when they are not supported by sufficient evidence.	Adapted from audit skepticism literature on questioning mindset and evidence sufficiency.
PS4	I pay close attention to inconsistencies between sustainability disclosures and supporting evidence.	Contextualized item targeting inconsistencies between ESG disclosure and underlying evidence.
PS5	I actively seek additional information when evaluating unfamiliar or complex sustainability disclosures.	Adapted from search-for-knowledge/questioning dimensions of skepticism and contextualized to complex sustainability disclosures.

Note: PS = professional skepticism.

- **Sustainability assurance self-efficacy**

Operational definition: Auditors' belief in their capability to understand requirements, evaluate evidence, adapt audit skills, and perform tasks related to sustainability assurance engagements.

Measurement basis: Developed as task-specific self-efficacy items based on self-efficacy theory and general self-efficacy measurement principles, then contextualized to sustainability assurance requirements, evidence evaluation, and auditor competency development (Alamin et al., 2020; Bandura, 1977; Chen et al., 2001; International Auditing & Assurance Standards Board, 2024).

Table A4. Measurement items for SASE

Code	Measurement Item	Item-Level Adaptation/Source Note
SASE1	I am confident that I can understand and apply relevant sustainability assurance requirements.	Task-specific capability item focused on understanding and applying sustainability assurance requirements.
SASE2	I am confident that I can evaluate evidence related to sustainability information using appropriate standards and guidance.	Task-specific capability item focused on ESG evidence evaluation under relevant standards/guidance.
SASE3	I can adapt my existing audit skills to sustainability assurance engagements.	Capability-transfer item linking traditional audit skills to sustainability assurance tasks.
SASE4	Even when sustainability assurance tasks are complex, I believe I can perform them effectively.	Self-efficacy item reflecting confidence under task complexity.

Note: SASE = sustainability assurance self-efficacy.

- **Attitude toward sustainability assurance**

Operational definition: Auditors' overall favorable or unfavorable evaluation of auditors' engagement in sustainability assurance as a professional activity.

Measurement basis: Adapted from the Theory of Planned Behavior attitude measurement logic and contextualized to sustainability assurance as an audit/professional service (Ajzen, 1991; Ajzen, 2020; Cohen & Simnett, 2015; Gipper et al., 2025).

Table A5. Measurement items for ATSA

Code	Measurement Item	Item-Level Adaptation/Source Note
ATSA1	I think auditors' engagement in sustainability assurance is beneficial for the audit profession.	Attitude item capturing perceived benefit for the audit profession.
ATSA2	I have a favorable view of auditors' involvement in sustainability assurance.	Direct favorable-evaluation item adapted from TPB attitude measurement logic.
ATSA3	I believe sustainability assurance is a valuable extension of traditional assurance services.	Attitude item contextualized to the expansion of traditional assurance services.
ATSA4	For auditors, engaging in sustainability assurance is professionally worthwhile.	Attitude item capturing perceived professional worthwhileness.

Note: ATSA = attitude toward sustainability assurance.

- **Intention to engage in sustainability assurance**

Operational definition: Auditors' motivational readiness, willingness, and stated plan to participate in sustainability assurance engagements when opportunities arise.

Measurement basis: Adapted from the Theory of Planned Behavior intention measurement logic and

contextualized to auditing and sustainability assurance engagement settings (Ajzen, 1991; Ajzen, 2020; Alamin et al., 2020; Alleyne et al., 2013; Tuan Mansor et al., 2021).

Table A6. Measurement items for intention to IESA

Code	Measurement Item	Item-Level Adaptation/Source Note
IESA1	I intend to participate in sustainability assurance engagements when opportunities arise.	Direct intention item adapted from Theory of Planned Behavior (TPB) behavioral intention measurement logic.
IESA2	I plan to seek opportunities to be involved in sustainability assurance work.	Planning-oriented intention item adapted from TPB intention logic.
IESA3	I would be willing to take part in sustainability assurance engagements if such opportunities are available in my firm.	Willingness/acceptance item contextualized to firm-level engagement opportunities; revised from assigned-task wording to reduce passivity.
IESA4	I intend to include sustainability assurance as part of my future professional practice.	Future professional-practice intention item contextualized to auditor role transformation.

Note: IESA = engage in sustainability assurance.